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Spruce budworm bacterial communities vary among sites and host tree species in a boreal landscape

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Handling Editor: Rosemary Gillespie**Abstract**

Aim: Microbial communities often vary spatially in how they assemble and knowledge is lacking about which factors determine the biogeography of host-associated microbiomes. Our aim is to assess the relative importance of spatial, environmental and host-associated factors on microbial community composition of an important defoliating insect.

Location: Boreal forests in eastern Canada (Quebec, Newfoundland and Labrador).

Taxon: The eastern spruce budworm (*Choristoneura fumiferana*) and its associated bacterial communities.

Methods: We characterized bacterial communities associated with spruce budworm larvae using bacterial 16S rRNA gene amplicon sequencing. We quantified how much of the variation in these bacterial communities could be explained by (1) environmental conditions, (2) the microbiome of foliage the larvae were eating, (3) host tree species and (4) spatial structure as quantified using Moran's Eigenvector Maps (MEMs).

Results: Budworm larval microbiomes varied significantly among sites and between host tree species. Larvae bacterial community structure was strongly correlated with the structure of bacterial communities taken from paired foliage samples. Spatial structure, foliage bacterial communities and host tree species collectively explained almost one-sixth of the variation in budworm bacterial communities while environmental conditions did not explain variation on their own.

Main conclusions: Lepidopteran microbiomes primarily originate from the foliage diet. However, subtle differences in microbial communities between larvae and foliage suggest that some bacteria establish and grow in the budworm microbiome, and that dispersal of bacteria from sources other than foliage as well as differences in environmental filtering between larval bodies and foliage play a role in the assembly of the budworm microbiome. While spatial location and spatial structure were also important drivers of spruce budworm bacterial community composition, none of the environmental variables we measured could explain the variation among sites, and identifying the drivers of this spatial variation remains an open question that will need to be addressed by future studies.

KEYWORDS

16S rRNA gene, defoliating insects, environmental predictors, lepidoptera, microbial dispersal limitation, microbiome, spatial variation

1 | INTRODUCTION

Insect microbiomes vary greatly both among species and geographically. Despite the numerous beneficial relationships having been found between insects and their microbiota (Breznak, 1982; Calderón-Cortés et al., 2011; Smith et al., 2015; Warnecke et al., 2007), knowledge of the processes governing the biogeography and assembly of insect microbiomes is lacking (Engel & Moran, 2013). Given the ecological importance of insects (Dimopoulos, 2003; Hung et al., 2018; Wright, 1986) and the importance of microbe-associated benefits on hosts and ecosystems, there is increasing interest in understanding the biogeography of insect microbiomes. Some studies have suggested that Lepidoptera may lack a persistent, resident gut bacterial community and that their microbiomes are instead populated mostly by transient bacteria that are derived from dietary sources (Hammer et al., 2017; Schwarz et al., 2018). One such important Lepidopteran forest pest is the eastern spruce budworm, *Choristoneura fumiferana*, which is one of the most destructive forest insects in North America (Royama, 1984) and is responsible for the defoliation of almost 13,500,000 hectares in the current outbreak in Québec (Ministère des Forêts, 2020).

The spruce budworm is part of a complex, multitrophic food web in which many other organisms affect its population and outbreak dynamics (Royama, 1984). The role of multitrophic interactions in spruce budworm outbreaks has been explored in detail for insect parasitoids (Eveleigh et al., 2007; Marrec et al., 2018) and avian predators (Crawford & Jennings, 1989; Germain et al., 2021). However, less attention has been given to the role of the spruce budworm microbiome, even though bacteria and fungi could play an important role in budworm population dynamics (Chen et al., 2016, 2020; Xia et al., 2013). In general, little is known about the factors that drive biogeographical variation in microbiome communities and how they respond to environmental gradients (Cohen et al., 2020), and this is particularly true for the spruce budworm (Landry et al., 2015). Improved understanding of what drives biogeographical patterns in spruce budworm-associated bacterial communities could lead to a better understanding of the drivers of budworm spatial population dynamics, which could be used to better understand outbreak mechanics (Broderick et al., 2009).

Biogeographical variation in budworm bacterial communities is influenced by the community of bacteria present in the insect's diet and those that are present in the insect's environment (Nemergut et al., 2013). The community present in the environment are filtered by the budworm host (Engel & Moran, 2013). Consequently, the pool of environmentally available microbes that colonize the budworm microbiome are likely to be determined by a variety of factors including environmental conditions, foliar bacterial communities, host tree species and other underlying spatial structure. Environmental conditions, including climate, affect the structure of ecological communities (Hernández-Gómez, 2020; Sepulveda & Moeller, 2020) by acting as filters that allow only a subset of the regional species pool

to disperse and establish in a habitat (Kraft et al., 2015). Ecological filters are a form of selection, that is, an ecological process by which organisms possessing specific traits are favoured in an environment (Nemergut et al., 2013; Vellend, 2016). Abiotic factors such as moisture, temperature and precipitation all affect bacterial growth (Balser et al., 2010). Climatic differences between sites can therefore modulate the selection and establishment of microbial populations in natural environments. The spatial configuration of populations of microbiome hosts can also affect microbiome community composition by affecting dispersal from a regional species pool (Nemergut et al., 2013; Vellend, 2016). Finally, spatial structure among sites can be used as a proxy for unmeasured spatial or environmental variation (Borcard et al., 2011) that may influence microbiome assembly, filtering or dispersal. Currently, we do not have a strong understanding of the relative importance of these different factors in determining community assembly and biogeography of insect-associated microbiomes.

An ongoing debate in microbial ecology has focused on whether animal-associated microbiomes are composed primarily of resident versus transient taxa (Astudillo-García et al., 2017; Hammer et al., 2019; Martinson et al., 2019). In insects, some studies have demonstrated that some microbes from the diet can establish and grow in the arthropod microbiome (Kennedy et al., 2020). Conversely, Hammer et al. (2017) argued that the Lepidopteran gut was devoid of a resident microbiome, that most bacteria are transient within the insect and that most bacteria are imported through diet. Previous work has shown that the bacterial communities residing on tree foliage vary spatially and among tree species (Laforest-Lapointe et al., 2016). If environmental and spatial variability impact foliar bacterial communities, and foliage is the spruce budworm's major dietary source of bacterial inoculants, foliar bacterial communities absorbed through foliage consumption should be the main determinant of the budworm bacterial community. Because the spruce budworm feeds on both spruce and balsam fir, bacterial community composition might also vary as a function of host tree species. The budworm itself could also act as an ecological filter and select for only certain bacterial taxa, in which case larval microbiomes could represent a subset of the foliage communities and specific bacterial taxa on the foliage could be consistently excluded by budworm body conditions.

In this study, we investigated what determines spatial variation in spruce budworm bacterial community composition. Our specific objective was to quantify the relative importance of four factors: (1) environmental conditions, (2) foliar bacterial communities, (3) host tree species and (4) spatial structure. We also sought to determine the degree to which these multiple factors interact to shape bacterial community composition. Because Lepidoptera are thought to lack a resident microbiome (Hammer et al., 2017), we hypothesized that the bacterial communities found on the foliage on which budworm feed should reflect the bacterial community found both in and on spruce budworm larvae. We also hypothesized that because foliar bacterial communities vary geographically and among host tree species (Laforest-Lapointe et al., 2016), spatial structure and host

tree species will be important drivers of microbial community structure in both budworm and foliage.

2 | MATERIALS AND METHODS

2.1 | Study area

We sampled spruce budworm bacterial communities and associated foliar bacterial communities at 26 sites along a gradient of outbreak density from Gaspésie to the Côte-Nord in Québec to Newfoundland & Labrador (Figure 1) in an area spanning more than 25,000 km². Our study area included sites near Baie-Comeau, Québec, which is the epicentre of the current budworm outbreak (Ministère des Forêts, 2020), as well as surrounding areas that had lower budworm population densities at the time of sampling. All sites (Table S1) were forested and contained both balsam fir (*Abies balsamea*) and black spruce (*Picea mariana*). For each site, we calculated mean annual temperature, total annual precipitation, solar radiation, evaporation, climatic moisture deficit and relative humidity using ClimateNA v5.10 (Wang et al., 2016).

2.2 | Sample collection

In the summer of 2015, we collected sixth instar larvae (the last larval stage of the spruce budworm) from all sites. At each site, we collected three larvae from three black spruce and three balsam fir randomly selected trees using aseptic techniques. We also collected a foliage sample consisting in a bud from each tree. Tree-level defoliation was visually estimated using the Fettes defoliation index (Fettes, 1951) to be used as an environmental factor. Larval and foliar samples were placed in sterile containers with RNAlater (Thermo Fisher, 2010) and stored on ice for approximately 2 days before being shipped to the laboratory, where they were stored at -80°C.

2.3 | Processing

We characterized the bacterial communities of whole insects at the level of individual trees. Collected larvae from each tree were pooled and homogenized together without any surface sterilization. Foliage samples were crushed by intensive bead-beating with 2.3 mm sterile steel beads in a BioSpec Mini-Beadbeater-16 prior to DNA extraction. DNA was extracted from larval and foliar samples using the Qiagen PowerSoil™ DNA extraction kit according to the manufacturer's instructions. Following DNA extraction, we amplified the bacterial 16S rRNA gene regions V5-V6 using the 799F-1193R primer pair (Chelius & Triplett, 2001) with barcodes and Illumina adapters. Amplified larval and foliar samples were concentration-equalized with the Invitrogen Sequalprep kit before paired-end sequencing with Illumina MiSeq V3 technology to obtain 2x300-nucleotide reads. Extraction negatives (water processed with the extraction kit) and PCR negatives were also sequenced and incorporated in the analysis.

Raw sequencing reads were demultiplexed with the *idemp* software (<https://github.com/yhwu/idemp/blob/master>) and then imported into the DADA2 1.12.1 package (Callahan et al., 2016) in R 3.6.0 (R Development Core Team, 2013) for processing. We filtered low-quality reads, merged paired-end reads and trimmed 20 nucleotides from the 5' end of each read to remove barcodes and primers. We then clustered reads into amplicon sequence variants (ASVs) using the DADA2 default parameters for all functions, and we annotated the taxonomic identity of each ASV by comparison with the SILVA database version 128 (Quast et al., 2012).

2.4 | Statistical analyses

We first used rarefaction curves to determine whether sequencing depth and sampling effort were adequate to capture a sufficient portion of bacterial diversity (Gotelli & Colwell, 2001). Samples with fewer than 4000 sequences were discarded, and the remaining

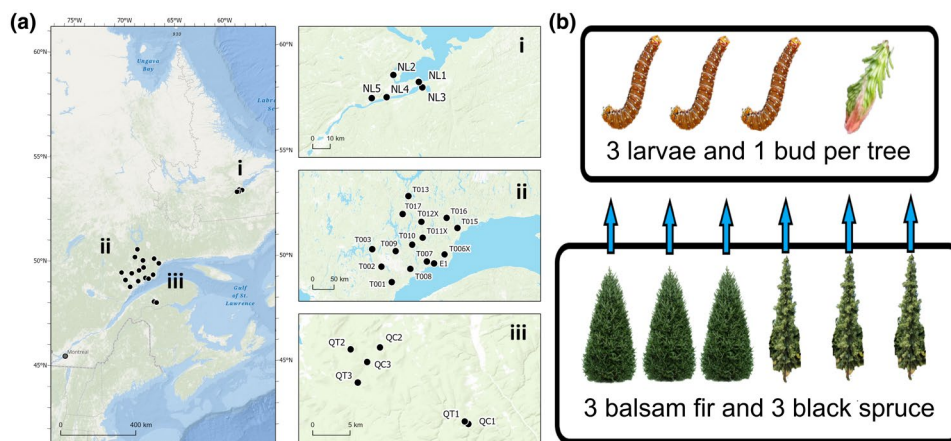


FIGURE 1 (a) Map of sampling sites in Québec, Canada and surrounding provinces. (b) Schematic of the sampling design at each site

samples were rarefied to 4000 sequences per sample; these criteria were sufficient to capture the majority of bacterial ASV diversity while retaining 95% of the samples (Figure S2). All subsequent analyses except for differential abundance analysis were performed on the rarefied data.

2.5 | Characterizing spatial structure

To include spatial predictors that account for unmeasured spatial variation, we quantified the spatial structure of our sampling sites using Moran's eigenvector maps (MEMs) (Borcard et al., 2011). MEMs are orthogonal vectors that quantify spatial structure in geospatial data (Legendre et al., 2015). These synthetic variables, referred to as MEM eigenvectors, capture spatial structure at different spatial scales and can be used as predictors in constrained ordination models (e.g. redundancy analysis [RDA]). The use of MEMs in an RDA framework to quantify spatial structure in ecological and genetic data is preferred relative to other methods often used in ecology, such as the Mantel test, due to its lower type 1 error rate (Harmon & Glor, 2010; Legendre et al., 2015; Lisboa et al., 2014).

2.6 | Larval and foliage community variation

We tested whether bacterial community composition varied significantly among geographical location and between host tree species (Anderson, 2017) using hierarchical modelling with permutational multivariate analyses of variance (PERMANOVAs) performed with the *adonis* function in *vegan* 2.5 in R. Bacterial taxa that were differentially abundant between spruce-feeding and fir-feeding budworm were then identified using the Analysis of Compositions of Microbiomes with Bias Correction (ANCOM-BC) method which models absolute abundance in a linear regression framework (Lin & Peddada, 2020). We used ANCOM-BC with default parameters, with two exceptions: (1) we filtered for taxa present in at least two samples and (2) we enabled structural zeros detection. We then used Principal Coordinates Analysis (PCoA) based on the Bray–Curtis dissimilarity between samples to visualize differences in community composition (Gower, 1966). To determine whether our measured environmental and spatial variables explained overall variation in bacterial community composition, we used the *envfit* function of the R package *vegan* 2.5 (Oksanen et al., 2016) to identify and plot predictors that were significantly correlated with the ordination axes ($p < 0.05$).

2.7 | Correlation between foliar and larval bacterial communities

The data used for comparing foliar and larval communities included all larval samples that could be associated their homologous

foliage sample with enough reads (>4000). We measured the relationship between foliar and larval bacterial communities using the Procrustes rotation test (Mardia et al., 1979; Peres-Neto & Jackson, 2001) using the PCoA-derived ordination solutions for both datasets as inputs. ANCOM-BC was used as before to identify taxa whose abundance was significantly different between larval and foliar samples.

2.8 | Modelling variation in bacterial community composition

We divided potential predictors of spruce budworm bacterial community composition into four categories to investigate the contribution of each predictor to community variation among samples while controlling for the effect of the others. The four categories were as follows: (1) environment, which included host tree damage (Fettes defoliation index), elevation and site-based climate variables; (2) foliar bacterial communities; (3) host tree species; and (4) spatial structure which was represented by MEMs. For this analysis, the bacterial community composition of foliage was represented using the scores of the first two axes of the PCoA ordination of the foliage bacterial communities. All continuous predictors were standardized to zero mean and unit variance prior to analysis. The response matrix, which in this case was the matrix of bacterial community relative abundance data, was Hellinger-transformed (Legendre & Gallagher, 2001).

We next identified the best predictors within each of the four categories described above by fitting models of community structure using redundancy analysis (RDA). For each category, we constructed an RDA model predicting bacterial community composition, and we reduced the set of potential predictors in the category using a bidirectional stepwise model selection procedure implemented with the *ordstep* function in *vegan*. Permutation p -values for each variable were used as the criterion for variable retention and model simplification. The variables that were retained in each of the four components after model simplification were as follows: (1) precipitation; (2) the scores of the first two axes of the foliar community PCoA ordination; (3) host tree species (i.e. fir or spruce); and (4) two of the eight eigenvectors obtained from the MEMs {1, 8}.

For our final model, we used the same transformed matrix indicated above as the response and the significant variables retained in each of the four categories as predictors. The unique and shared contributions of each predictor category to overall biogeographical variation in bacterial community structure among budworm larvae were then assessed using variance partitioning (Borcard et al., 1992). It is important to note that predictors that were excluded from this final analysis may still potentially impact budworm communities; they were removed during stepwise selection on the basis of permutational p -values, but this does not imply that they are necessarily unimportant. Correlations among predictors are shown in Tables S4 and S5.

3 | RESULTS

3.1 | Larval and foliage community variation

Spruce budworm bacterial communities were dominated by the phyla *Firmicutes* ($39.6 \pm 4.0\%$ relative abundance) and *Proteobacteria* ($36.3 \pm 3.6\%$), whereas foliage communities were mostly dominated by *Proteobacteria* ($53.8 \pm 1.5\%$). We detected a total of 131 ASVs in larvae and 115 ASVs in foliage. Because negative control samples were either very different from experimental samples or had very few reads, indicating that reads in experimental samples represented budworm or foliage bacteria, they were removed from the analysis. The cut-off of 4,000 reads per larval and foliar samples was sufficient to ensure that all ASVs represented all bacterial diversity in our samples, as shown by the asymptotic rarefaction curve (Figure S1). The average ($\pm$ SD) number of ASVs per sample was 24 ± 11 for larvae and 28 ± 11 for foliage. There was no significant difference in the average number of ASVs found in larval samples versus foliage samples (ANOVA; $F = 2.437$, $p = 0.121$).

Region and sample site explained almost one-third of the variation in larval bacterial community composition (PERMANOVA; Region: $R^2 = 0.04$, $p = 0.001$, Site: $R^2 = 0.25$, $p = 0.001$), indicating that insects at each site had a distinctive bacterial community structure (Table 1) and that most of spatial variation was between individual sites instead of between regions. Host tree species nested within sample site explained an additional 21% of the variation in community structure (PERMANOVA $R^2 = 0.21$, $p = 0.026$). At

deeper phylogenetic levels such as Genus and Family, the variation explained by site and region stayed the same while host tree species became an insignificant term in the model (Table 1). Differential abundance analysis detected ASVs that were positively associated with larvae that fed on balsam fir, which were mostly *Firmicutes* and *Proteobacteria*, specifically *Acetobacter* and *Pseudomonas* genera (Figure 2). When we overlaid environmental vectors on ordinations of community composition (Figure 3), we found that both climate factors and MEMs 1, 5 and 8 were significantly correlated with both ordination axes. When looking at which spatial patterns these MEMs represented, MEM1 appeared to capture the spatial variation among the three regions, while MEMs 5 and 8 captured uncharacterized fine-scale spatial variation (Figure S2).

3.2 | Modelling variation in bacterial community structure

We used variance partitioning to quantify the unique and shared contributions of environmental variation, foliar bacterial communities, host tree species and spatial structure to budworm bacterial community composition. Collectively, all four components explained 16% of the overall variation in spruce budworm bacterial community composition (Figure 4, Table S6). Of the four components, spatial structure and foliage bacterial communities best explained variation in spruce budworm larval microbiome composition among sites; spatial structure explained a unique 5% of this variation and foliage

TABLE 1 Variation in spruce budworm bacterial communities as a function of sampling site and host tree species, tested using a Bray–Curtis dissimilarity-based PERMANOVA on ASV, genus- and family-level community tables. Host tree species was nested within site which was nested within region for this analysis

	Df	Sum Sq	Mean Sq	F-Value	R^2	p-Value
ASV						
Region	2	1.913	0.95629	2.5538	0.04435	0.001
Site (within region)	23	10.703	0.46535	1.2427	0.24821	0.001
Tree species (within site)	22	9.162	0.41644	1.1121	0.21246	0.026
Residuals	57	21.344	0.37446		0.49498	
Total	104	43.122			1	
Genus						
Region	2	2.116	1.01818	3.6888	0.06254	0.001
Site (within region)	23	8.53	0.37089	1.2929	0.2521	0.002
Tree species (within site)	22	6.84	0.31089	1.0838	0.20213	0.202
Residuals	57	16.351	0.28686		0.48322	
Total	104	33.837			1	
Family						
Region	2	2.1575	1.07876	4.2325	0.07016	0.001
Site (within region)	23	8.0207	0.34872	1.3682	0.26084	0.003
Tree species (within site)	22	6.044	0.27473	1.0779	0.19655	0.231
Residuals	57	14.5277	0.25487		0.47245	
Total	104	30.7499			1	

Values in bold indicate statistically significant predictors of spruce budworm bacterial community composition ($p < 0.05$).

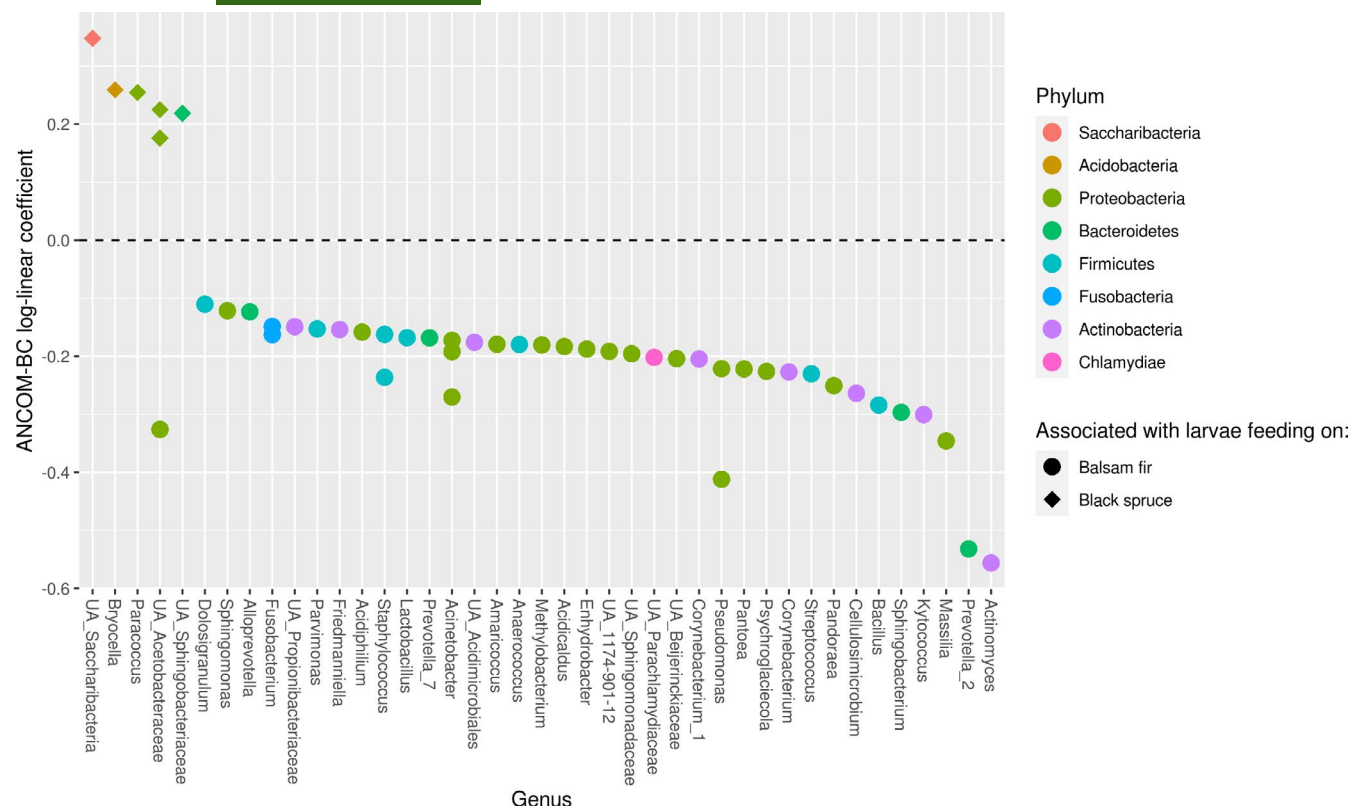


FIGURE 2 ANCOM-BC discriminant analysis of differentially abundant ASVs in spruce budworm larvae coming from different host tree species. Most differentially abundant ASVs were positively associated with balsam fir and were from the *Proteobacteria* and *Actinobacteria* phyla

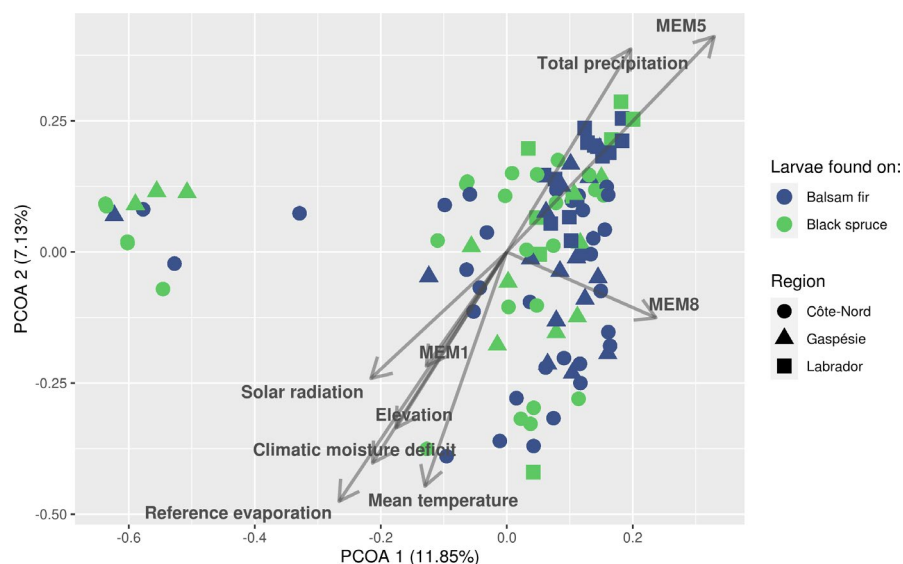
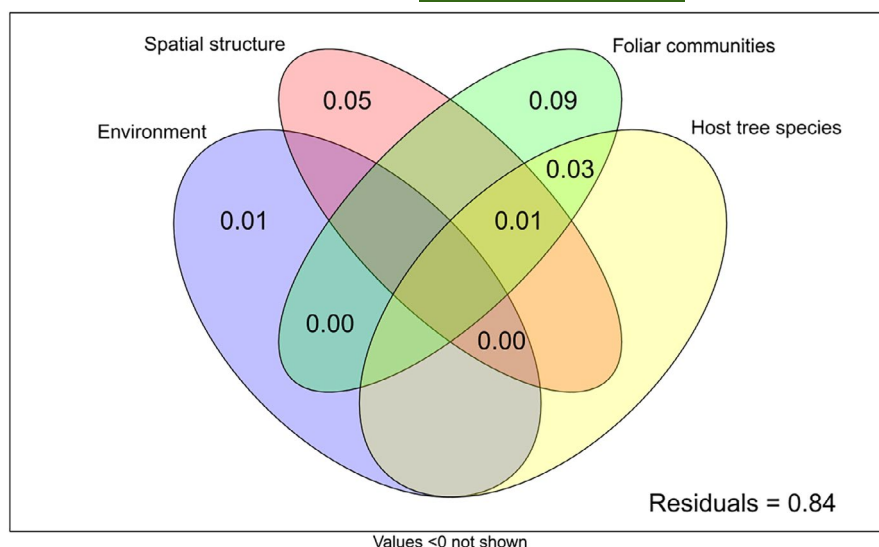


FIGURE 3 Principal coordinates (PCoA) ordination based on the Bray-Curtis dissimilarity between bacterial communities, Vectors indicate variables that were significantly correlated with ordination axes ($p < 0.05$) based on a randomization test. Variance explained by each PCoA axis is shown in axis labels. Most predictor variables clustered into two separate groups of intercorrelated vectors that were correlated with the two ordination axes, indicating that the composition of spruce budworm bacterial communities covaried with several measured environmental variables

bacterial communities explained another unique 9%. Host tree species had shared components with spatial structure and foliage communities that explained 1% and 3% of the variation, respectively.

When larval bacterial communities were analysed separately for the two host tree species, we found that different environmental factors were correlated with the first two ordination axes for each host tree (Figure S3). Spatial structure (MEMs) and environmental

factors including precipitation, evaporation, temperature, moisture and solar radiation were more strongly associated with bacterial community structure in larvae feeding on balsam fir than in larvae feeding on black spruce. PCoA axes for bacterial communities found in larvae feeding on black spruce were only correlated with the spatial structure variable MEM5, which captured fine-scale spatial variation, elevation and evaporation (Table S2).



abundant between larval and foliage communities: *Actinobacteria* were more abundant in larval samples and *Proteobacteria* were more abundant in foliar samples (Figure 5). Like larval communities, variation in foliar bacterial communities was significantly associated with geographical location (i.e. site), which explained 71% of the variance among samples (PERMANOVA; $R^2 = 0.71$, $p = 0.044$).

4 | DISCUSSION

We assessed how environmental factors, host tree species, foliar communities and spatial structure affected the biogeography of microbes inhabiting the spruce budworm. We accomplished this by characterizing the bacterial communities of budworm larvae collected on two different host tree species at 26 different sampling sites in eastern Canada. We specifically sought to test the hypothesis that budworm-associated bacterial communities are determined mainly by the bacterial communities on the foliage that they consume. We confirmed this hypothesis in identifying a significant correlation between the ordination solutions for both communities (Procrustes test; $R^2 = 0.64$, $p = 0.001$), and given the proportion of variance in larval community structure explained by foliar communities (Figure 4). We also confirmed our hypothesis that host tree species and spatial structure affect budworm bacterial communities.

We identified significant biogeographical variation in spruce budworm bacterial community structure among the 26 sampling sites spanning more than 800 kilometres. Most of this variation was among sites (25%) and between different host tree species within sites (22%; Table 1). A small portion of the variation was also explained by region (4%) showing that most of the spatial variation in budworm communities was present at a finer spatial scale than region. Among-site variation could be linked to uncharacterized fine-scale spatial variation such as unmeasured environmental variation or further variation among foliar bacterial communities. Indeed, we identified a significant contribution of fine-scale MEMs that could reflect these hypothesized processes. Previous work has also found that foliar bacterial community structure is highly variable among sites (Laforest-Lapointe et al., 2016).

Strong among-site differences in community composition suggest that dispersal limitation or filtering by spatially autocorrelated environmental factors are driving budworm bacterial community composition. Despite these site-level differences, none of the environmental or climatic variables we measured explained this variation (Figure 3). Instead, MEMs that capture spatial structure, but do not identify its root cause, were among the best predictors of spruce budworm bacterial community structure. As a result, the biological factors underlying this spatial variation in community structure remain unknown. The fine- and coarse-scale MEMs that explained community structure likely capture unmeasured climatic factors or dispersal limitation of the budworm, their associated bacterial communities or both. Previous studies have shown that climatic conditions have a large influence on moth dispersal (Halsch et al., 2021) and could impact budworm distribution (Régnière et al., 2012). However, because of the similarity between foliar and larval communities, the spatial variation we measured is likely to also be influenced by free-living bacteria dispersal captured by larval communities in addition to bacteria being transported by budworm larvae. Studies have shown that ASV-based bacterial analyses can overestimate the importance of dispersal limitation (Raulo et al., 2021; Stothart et al., 2021). For this reason, we repeated our analyses of community variation explained by sampling site at different phylogenetic levels

(Table 1). The results were similar, indicating that dispersal limitation is relatively constant whether ASVs are aggregated at the family or genus level and that the dispersal traits of budworm-associated bacterial communities could be conserved at deeper phylogenetic level (Martiny et al., 2013, 2015).

Host tree species had a measurable impact on both larval and foliar bacterial communities. More than 22% of the overall variance in bacterial community structure was explained by the identity of host tree species. With respect to the determinants of these differences, we found that budworm bacterial communities from different host tree species were correlated with different explanatory variables; for example, most environmental conditions had an impact on the bacterial communities of larvae on balsam fir but not on those on black spruce (Table S3). The effect of host tree species on spruce budworm larval bacterial communities could be explained by differences in the chemical composition of the foliage of the two host tree species and which would drive differences in foliage-associated bacteria that are then ingested by the insect. Animals who are a source of bacterial dispersal on foliage could also prefer one tree species over another (Beltrán & Wunderle, 2013), impacting the available dispersal pool to the budworm on different host tree species. Additionally, conifer needles have been shown to generally inhibit bacterial growth in the spruce budworm due to the antibacterial effect of tannins (Pang, 2010), and this effect could vary between tree species.

Overall, we found that microbial communities on the budworm's foliage diet were the most important driver of budworm microbiome composition. This result suggests that the key process determining bacterial community structure within the spruce budworm is colonization by microbes from their diet, and is consistent with the hypothesis that Lepidoptera lack a resident microbiome (Hammer et al., 2017; Schwarz et al., 2018), which predicts that larvae should have the same bacterial community as that found on their foliage diet. Our demonstration of the broad similarities between the microbiomes of the spruce budworm and its foliage diet provides some of the first field-based evidence to support the hypothesis that some Lepidoptera lack a persistent resident microbiome. However, there were some differences in taxonomic composition between the spruce budworm and foliage microbiota and some taxa that were associated with the budworm microbiome, and these differences likely represent taxa that are able to establish and grow in the budworm microbiome, as has been observed in other arthropod-associated microbial communities (Kennedy et al., 2020); for example, *Actinobacteria* were significantly more abundant in larvae as were several ASVs from diverse phyla, whereas *Proteobacteria* were more abundant in foliage. The *Actinobacteria* and other taxa associated with larvae but not foliage may originate elsewhere in the larvae's environment, or from vertical transmission. These taxa could also be selectively enriched from species in the foliage-associated communities that were too rare to be detected, due to selection by the biotic or abiotic conditions in the budworm microbiome leading to strong increases in abundance and increased possibility of detection. Thus, while the majority of the diversity in the budworm microbiome is likely transient taxa originating from foliage, some bacterial taxa



do appear to be likely residents of the microbiome. Foliar communities are themselves impacted by environmental factors including air quality and climate (Espenshade et al., 2019; Laforest-Lapointe et al., 2016; Magan & McLeod, 1991), suggesting that future studies that take into account direct and indirect environmental drivers of animal microbiomes will be required to understand the full spectrum of factors driving the biogeography of these microbiomes.

The spatial and environmental factors we measured explained almost a sixth (16%) of variance in community composition, with the variance explained by foliage communities peaking at 9%. The poor explanatory power of our measured variables could be due to the importance of stochastic processes in budworm microbiome community assembly and biogeography, such as dispersal of insect hosts and microbes and subsequent colonization by microbes. The scale at which we quantified spatial structure may not capture well such dispersal and colonization processes. Genetic variation in budworms or host trees could be a potential explanation for the important amount of spatial variation in budworm bacterial communities yet to be explained. Spruce budworm spatial genetic structure in our study area (Côte-Nord and Gaspésie, Figure 1) was generally homogeneous, with some subtle genetic differences between the two regions (Larroque et al., 2020). Because most of the inter-site bacterial community variation was at a finer spatial scale than that of these regional budworm differences (Table 1), it seems likely that the impact of budworm genetic variation on bacterial communities is minimal. However, as foliar communities are closely linked to larval communities, linking host tree genetic variation in the outbreak area to budworm bacterial communities could be a promising avenue of future research.

5 | CONCLUSIONS

We have shown that the community assembly and biogeography of spruce budworm bacterial communities are shaped by spatial structure among sites, foliage communities and host tree species. Most of the variation in budworm communities was mirrored in the communities of foliage they ate, showing that foliage bacterial communities are likely the main determinant of budworm microbiome composition. While spatial location (i.e. site) and spatial structure of sites were important indicators of spruce budworm bacterial community composition, none of the environmental variables we measured could explain the variation among sites, and identifying the drivers of this spatial variation remains an open question that will need to be addressed by future studies. We also determined that host tree species is an important determinant of both spruce budworm microbiome composition and the sensitivity of that microbiome to the environment, as the microbiomes of budworms living on balsam fir were more strongly correlated with other spatial and environmental predictors. Finally, we found support for the hypothesis that Lepidopteran microbiomes are mostly non-resident and likely originate from the foliage diet. However, subtle differences in microbial communities between larvae and

foliage suggest that some bacteria do establish and grow in the budworm microbiome, and that dispersal of bacteria from other sources than foliage and differences in environmental filtering between larval bodies and foliage play a role in the assembly of the budworm microbiome. Our results advance understanding of the factors that drive the assembly and biogeography of bacterial communities associated with an ecologically and economically important forest insect, and open the door to understanding how these factors impact the biogeography of the budworm itself and its outbreak consequences.

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CONFLICT OF INTEREST

The authors declare that they have no conflict of interest.

DATA AVAILABILITY STATEMENT

The datasets analysed as part of this study are available in a Genbank repository: PRJNA701313. The scripts used to perform the analyses for this study are available in a GitHub repository: https://github.com/mathlandry/budworm_biogeo

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REFERENCES

- Anderson, M. J. (2017). Permutational multivariate analysis of variance (PERMANOVA). In *Wiley StatsRef: Statistics reference online* (pp. 1–15). John Wiley & Sons, Ltd. <https://doi.org/10.1002/9781118445112.stat07841>
- Astudillo-García, C., Bell, J. J., Webster, N. S., Glasl, B., Jompa, J., Montoya, J. M., & Taylor, M. W. (2017). Evaluating the core microbiota in complex communities: A systematic investigation. *Environmental Microbiology*, 19(4), 1450–1462. <https://doi.org/10.1111/1462-2920.13647>
- Balser, T. C., Gutknecht, J. L. M., & Liang, C. (2010). How will climate change impact soil microbial communities? In *Soil microbiology and sustainable crop production* (pp. 373–397). Springer Netherlands.
- Beltrán, W., & Wunderle, J. M. (2013). Determinants of tree species preference for foraging by insectivorous birds in a novel *Prosopis-Leucaena* woodland in Puerto Rico: The role of foliage palatability. *Biodiversity and Conservation*, 22(9), 2071–2089. <https://doi.org/10.1007/s10531-013-0529-x>
- Borcard, D., Gillet, F., & Legendre, P. (2011). *Numerical ecology with R*. Springer, New York. https://doi.org/10.1007/978-1-4419-7976-6_1
- Borcard, D., Legendre, P., & Drapeau, P. (1992). Partialling out the spatial component of ecological variation. *Ecology*, 73(3), 1045–1055. <https://doi.org/10.2307/1940179>
- Breznak, J. a (1982). Intestinal microbiota of termites and other xylophagous insects. *Annual Review of Microbiology*, 36, 323–343. <https://doi.org/10.1146/annurev.mi.36.100182.001543>
- Broderick, N. A., Robinson, C. J., McMahon, M. D., Holt, J., Handelsman, J., & Raffa, K. F. (2009). Contributions of gut bacteria to *Bacillus*

- thuringiensis*-induced mortality vary across a range of Lepidoptera. *BMC Biology*, 7(1), 11. <https://doi.org/10.1186/1741-7007-7-11>
- Calderón-Cortés, N., Quesada, M., Watanabe, H., Cano-Camacho, H., & Oyama, K. (2011). Endogenous plant cell wall digestion: A key mechanism in insect evolution. *Annual Review of Ecology, Evolution, and Systematics*, 43(1), 120830113150004. <https://doi.org/10.1146/annurev-ecolsys-110411-160312>
- Callahan, B. J., McMurdie, P. J., Rosen, M. J., Han, A. W., Johnson, A. J. A., & Holmes, S. P. (2016). DADA2: High-resolution sample inference from Illumina amplicon data. *Nature Methods*, 13(7), 581–583. <https://doi.org/10.1038/nmeth.3869>
- Chelius, M. K., & Triplett, E. W. (2001). The diversity of archaea and bacteria in association with the roots of *Zea mays* L. *Microbial Ecology*, 41(3), 252–263. <https://doi.org/10.1007/s002480000087>
- Chen, B., Teh, B. S., Sun, C., Hu, S., Lu, X., Boland, W., & Shao, Y. (2016). Biodiversity and activity of the gut microbiota across the life history of the insect herbivore *Spodoptera littoralis*. *Scientific Reports*, 6, 1–14. <https://doi.org/10.1038/srep29505>
- Chen, B., Zhang, N., Xie, S., Zhang, X., He, J., Muhammad, A., Sun, C., Lu, X., & Shao, Y. (2020). Gut bacteria of the silkworm *Bombyx mori* facilitate host resistance against the toxic effects of organophosphate insecticides. *Environment International*, 143, <https://doi.org/10.1016/j.envint.2020.105886>
- Cohen, H., McFrederick, Q. S., & Philpott, S. M. (2020). Environment shapes the microbiome of the blue orchard bee, *Osmia lignaria*: RRH: Environmental drivers of bee microbiome. *Microbial Ecology*, 80(4), 897–907. <https://doi.org/10.1007/s00248-020-01549-y>
- Crawford, H. S., & Jennings, D. T. (1989). Predation by birds on spruce budworm *Choristoneura fumiferana*: Functional numerical, and total responses. *Ecology*, 70(1), 152–163. <https://doi.org/10.2307/1938422>
- Dimopoulos, G. (2003). Insect immunity and its implication in mosquito-malaria interactions. In *Cellular microbiology* (Vol. 5, Issue 1, pp. 3–14). John Wiley & Sons, Ltd. <https://doi.org/10.1046/j.1462-5822.2003.00252.x>
- Engel, P., & Moran, N. A. (2013). The gut microbiota of insects—diversity in structure and function. *FEMS Microbiology Reviews*, 37(5), 699–735. <https://doi.org/10.1111/1574-6976.12025>
- Espenshade, J., Thijs, S., Gawronski, S., Bové, H., Weyens, N., & Vangronsveld, J. (2019). Influence of urbanization on epiphytic bacterial communities of the platanus × hispanica tree leaves in a biennial study. *Frontiers in Microbiology*, 10, 1–14. <https://doi.org/10.3389/fmicb.2019.00675>
- Eveleigh, E. S., McCann, K. S., McCarthy, P. C., Pollock, S. J., Lucarotti, C. J., Morin, B., McDougall, G. A., Strongman, D. B., Huber, J. T., Umbanhowar, J., & Faria, L. D. B. (2007). Fluctuations in density of an outbreak species drive diversity cascades in food webs. *Proceedings of the National Academy of Sciences of the United States of America*, 104, 16976–16981. <https://doi.org/10.1073/pnas.0704301104>
- Fettes, J. P. (1951). *Investigations of sampling techniques for population studies of the spruce budworm on balsam fir in Ontario*. <http://www.worldcat.org/title/investigations-of-sampling-techniques-for-population-studies-of-the-spruce-budworm-on-balsam-fir-in-ontario/oclc/62428296>
- Fisher, T. (2010). *RNA later preserves bacterial gene expression profiles for array analysis* - JP. <https://www.thermofisher.com/ca/en/home/references/ambion-tech-support/rna-isolation/techniques/rnalater-preserves-bacterial-gene-expression-profiles-for-array-.html>
- Germain, M., Kneeshaw, D., De Grandpré, L., Desrochers, M., James, P. M. A., Vepakomma, U., Poulin, J. F., & Villard, M. A. (2021). Insectivorous songbirds as early indicators of future defoliation by spruce budworm. *Landscape Ecology*, 36(10), 3013–3027. <https://doi.org/10.1007/s10980-021-01300-z>
- Gotelli, N. J., & Colwell, R. K. (2001). Quantifying biodiversity: Procedures and pitfalls in the measurement and comparison of species richness. In *Ecology letters* (Vol. 4, Issue 4, pp. 379–391). John Wiley & Sons, Ltd. <https://doi.org/10.1046/j.1461-0248.2001.00230.x>
- Gower, J. C. (1966). Some distance properties of latent root and vector methods used in multivariate analysis. *Biometrika*, 53(3/4), 325. <https://doi.org/10.2307/2333639>
- Halsch, C. A., Shapiro, A. M., Fordyce, J. A., Nice, C. C., Thorne, J. H., Waetjen, D. P., & Forister, M. L. (2021). Insects and recent climate change. *Proceedings of the National Academy of Sciences of the United States of America*, 118(2), 1–9. <https://doi.org/10.1073/PNAS.2002543117>
- Hammer, T. J., Janzen, D. H., Hallwachs, W., Jaffe, S. P., & Fierer, N. (2017). Caterpillars lack a resident gut microbiome. *Proceedings of the National Academy of Sciences of the United States of America*, 114(36), 9641–9646. <https://doi.org/10.1073/pnas.1707186114>
- Hammer, T. J., Sanders, J. G., & Fierer, N. (2019). Not all animals need a microbiome. In *FEMS microbiology letters* (Vol. 366, Issue 10, p. 117). Oxford Academic. <https://doi.org/10.1093/femsle/fnz117>
- Harmon, L. J., & Glor, R. E. (2010). Poor statistical performance of the mantel test in phylogenetic comparative analyses. *Evolution*, 64(7), 2173–2178. <https://doi.org/10.1111/j.1558-5646.2010.00973.x>
- Hernández-Gómez, O. (2020). Climate change disturbs wildlife microbiomes. *Nature Climate Change*, 10(11), 981–982. <https://doi.org/10.1038/s41558-020-00923-5>
- Hung, K.-L.-J., Kingston, J. M., Albrecht, M., Holway, D. A., & Kohn, J. R. (2018). The worldwide importance of honey bees as pollinators in natural habitats. *Proceedings of the Royal Society B: Biological Sciences*, 285(1870), 20172140. <https://doi.org/10.1098/rspb.2017.2140>
- Kennedy, S. R., Tsau, S., Gillespie, R., & Krehenwinkel, H. (2020). Are you what you eat? A highly transient and prey-influenced gut microbiome in the grey house spider *Badumna longinqua*. *Molecular Ecology*, 29(5), 1001–1015. <https://doi.org/10.1111/mec.15370>
- Kraft, N. J. B., Adler, P. B., Godoy, O., James, E. C., Fuller, S., & Levine, J. M. (2015). Community assembly, coexistence and the environmental filtering metaphor. *Functional Ecology*, 29(5), 592–599. <https://doi.org/10.1111/1365-2435.12345>
- Laforest-Lapointe, I., Messier, C., & Kembel, S. W. (2016). Host species identity, site and time drive temperate tree phyllosphere bacterial community structure. *Microbiome*, 4(1), 27. <https://doi.org/10.1186/s40168-016-0174-1>
- Landry, M., Comeau, A. M., Derome, N., Cusson, M., & Levesque, R. C. (2015). Composition of the spruce budworm (*Choristoneura fumiferana*) midgut microbiota as affected by rearing conditions. *PLoS One*, 10(12), e0144077. <https://doi.org/10.1371/journal.pone.0144077>
- Larroque, J., Johns, R., Canape, J., Morin, B., & James, P. M. A. (2020). Spatial genetic structure at the leading edge of a spruce budworm outbreak: The role of dispersal in outbreak spread. *Forest Ecology and Management*, 461, 117965. <https://doi.org/10.1016/j.foreco.2020.117965>
- Legendre, P., Fortin, M. J., & Borcard, D. (2015). Should the Mantel test be used in spatial analysis? *Methods in Ecology and Evolution*, 6(11), 1239–1247. <https://doi.org/10.1111/2041-210X.12425>
- Legendre, P., & Gallagher, E. D. (2001). Ecologically meaningful transformations for ordination of species data. *Oecologia*, 129(2), 271–280. <https://doi.org/10.1007/s004420100716>
- Lin, H., & Peddada, S. D. (2020). Analysis of compositions of microbiomes with bias correction. *Nature Communications*, 11(1), 1–11. <https://doi.org/10.1038/s41467-020-17041-7>
- Lisboa, F. J. G., Peres-Neto, P. R., Chaer, G. M., Jesus, E. D. C., Mitchell, R. J., Chapman, S. J., & Berbara, R. L. L. (2014). Much beyond Mantel: bringing procrustes association metric to the plant and soil ecologist's toolbox. *PLoS One*, 9(6), e101238. <https://doi.org/10.1371/journal.pone.0101238>
- Magan, N., & McLeod, A. R. (1991). *Effects of atmospheric pollutants on phyllosphere microbial communities* (pp. 379–400). Springer. https://doi.org/10.1007/978-1-4612-3168-4_19
- Mardia, K. V., Kent, J. T., Bibby, J. M. (1979). *Multivariate analysis*. Academic Press.



- Marrec, R., Pontbriand-Pare, O., Legault, S., & James, P. M. A. (2018). Spatiotemporal variation in drivers of parasitoid metacommunity structure in continuous forest landscapes. *Ecosphere*, 9(1), <https://doi.org/10.1002/ecs2.2075>
- Martinson, J. N. V., Pinkham, N. V., Peters, G. W., Cho, H., Heng, J., Rauch, M., Broadaway, S. C., & Walk, S. T. (2019). Rethinking gut microbiome residency and the Enterobacteriaceae in healthy human adults. *ISME Journal*, 13(9), 2306–2318. <https://doi.org/10.1038/s41396-019-0435-7>
- Martiny, A. C., Treseder, K., & Pusch, G. (2013). Phylogenetic conservatism of functional traits in microorganisms. *ISME Journal*, 7(4), 830–838. <https://doi.org/10.1038/ismej.2012.160>
- Martiny, J. B. H., Jones, S. E., Lennon, J. T., & Martiny, A. C. (2015). Microbiomes in light of traits: A phylogenetic perspective. *Science*, 350(6261), <https://doi.org/10.1126/science.aac9323>
- Ministère des Forêts, de la F. et des P. (2020). *Aires infestées par la tordeuse des bourgeons de l'épinette au Québec en 2020*. <http://nfdp.ccfm.org/fr/data/insects.php>
- Nemergut, D. R., Schmidt, S. K., Fukami, T., O'Neill, S. P., Bilinski, T. M., Stanish, L. F., Knelman, J. E., Darcy, J. L., Lynch, R. C., Wickey, P., & Ferrenberg, S. (2013). Patterns and processes of microbial community assembly. *Microbiology and Molecular Biology Reviews*, 77(3), 342–356. <https://doi.org/10.1128/MMBR.00051-12>
- Oksanen, J., Blanchet, F., Kindt, R., Legendre, P., Minchin, P., O'Hara, R. B., Simpson, G., Solymos, P., Stevens, M., & Wagner, H. (2016). *vegan: Community ecology R package*. <https://cran.r-project.org/package=vegan>
- Pang, A. S. D. (2010). Possible role of plant allelochemical in clearance of bacteria from the gut of spruce budworm, *Choristoneura fumiferana*. *The Open Microbiology Journal*, 4, 26–29. <https://doi.org/10.2174/1874285801004010026>
- Peres-Neto, P. R., & Jackson, D. A. (2001). How well do multivariate data sets match? The advantages of a procrustean superimposition approach over the Mantel test. *Oecologia*, 129(2), 169–178. <https://doi.org/10.1007/s004420100720>
- Quast, C., Pruesse, E., Yilmaz, P., Gerken, J., Schweer, T., Yarza, P., Peplies, J., & Glöckner, F. O. (2012). The SILVA ribosomal RNA gene database project: Improved data processing and web-based tools. *Nucleic Acids Research*, 41(D1), D590–D596. <https://doi.org/10.1093/nar/gks1219>
- R Development Core Team. (2013). *R: A language and environment for statistical computing*. R Foundation for Statistical Computing. <http://www.r-project.org/>
- Raulo, A., Allen, B. E., Troitsky, T., Husby, A., Firth, J. A., Coulson, T., & Knowles, S. C. L. (2021). Social networks strongly predict the gut microbiota of wild mice. *ISME Journal*, 15(9), 2601–2613. <https://doi.org/10.1038/s41396-021-00949-3>
- Régnière, J., St-Amant, R., & Duval, P. (2012). Predicting insect distributions under climate change from physiological responses: Spruce budworm as an example. *Biological Invasions*, 14(8), 1571–1586. <https://doi.org/10.1007/s10530-010-9918-1>
- Royama, T. (1984). Population dynamics of the spruce budworm *Choristoneura fumiferana*. *Ecological Monographs*, 54(4), 429–462. <https://doi.org/10.2307/1942595>
- Schwarz, M. T., Kneeshaw, D., & Kembel, S. W. (2018). The gut-associated microbiome of the eastern spruce budworm does not influence larval growth or survival. *BioRxiv*, 330928, <https://doi.org/10.1101/330928>
- Sepulveda, J., & Moeller, A. H. (2020). The effects of temperature on animal gut microbiomes. *Frontiers in Microbiology*, 11, 1–9. <https://doi.org/10.3389/fmicb.2020.00384>
- Smith, A. H., Łukasik, P., O'Connor, M. P., Lee, A., Mayo, G., Drott, M. T., Doll, S., Tuttle, R., Disciullo, R. A., Messina, A., Oliver, K. M., & Russell, J. A. (2015). Patterns, causes and consequences of defensive microbiome dynamics across multiple scales. *Molecular Ecology*, 24(5), 1135–1149. <https://doi.org/10.1111/mec.13095>
- Stothart, M. R., Greuel, R. J., Gavriluc, S., Henry, A., Wilson, A. J., McLoughlin, P. D., & Poissant, J. (2021). Bacterial dispersal and drift drive microbiome diversity patterns within a population of feral hindgut fermenters. *Molecular Ecology*, 30(2), 555–571. <https://doi.org/10.1111/mec.15747>
- Vellend, M. (2016). *The theory of ecological communities—Full book*. <https://press.princeton.edu/titles/10914.html>
- Wang, T., Hamann, A., Spittlehouse, D., & Carroll, C. (2016). Locally downscaled and spatially customizable climate data for historical and future periods for North America. *PLoS One*, 11(6), e0156720. <https://doi.org/10.1371/journal.pone.0156720>
- Warnecke, F., Luginbühl, P., Ivanova, N., Ghassemian, M., Richardson, T. H., Stege, J. T., Cayouette, M., McHardy, A. C., Djordjevic, G., Aboushadi, N., Sorek, R., Tringe, S. G., Podar, M., Martin, H. G., Kunin, V., Dalevi, D., Madejska, J., Kirton, E., Platt, D., ... Leadbetter, J. R. (2007). Metagenomic and functional analysis of hindgut microbiota of a wood-feeding higher termite. *Nature*, 450, 560–565. <https://doi.org/10.1038/nature06269>
- Wright, D. E. (1986). Damage and loss in yield of wheat crops caused by the Australian plague locust, *Chortoicetes terminifera* (Walker). *Australian Journal of Experimental Agriculture*, 26(5), 613–618. <https://doi.org/10.1071/EA9860613>
- Xia, X., Zheng, D., Zhong, H., Qin, B., Gurr, G. M., Vasseur, L., Lin, H., Bai, J., He, W., & You, M. (2013). DNA sequencing reveals the midgut microbiota of diamondback moth, *Plutella xylostella* (L.) and a possible relationship with insecticide resistance. *PLoS One*, 8(7), <https://doi.org/10.1371/journal.pone.0068852>

BIOSKETCH

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Author contribution: ML carried out the data collection, sample processing, data analysis and drafting the article. ML, PJ, DK and SK all contributed to conception of the work, data interpretation, critical revision of the article and final approval of the version to be published.

SUPPORTING INFORMATION

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